



# SHASHWAT TIWARI

## DATA SCIENTIST

Data Science professional with close to 5 years of work experience in Machine Learning, Deep Learning and Big Data. Currently working as **Data Scientist** with **EY**, Noida, India. Strong expertise in Machine Learning and Deep learning in the areas of Linear & Logistic Regression, Decision trees, Ensemble Learning, Clustering, CNN, LSTM, RNN and GANs. Shashwat is also well versed with **Python, SQL & PySpark**.

## 🔑 SUMMARY OF MAIN SKILLS

### **Programming**

#### **Languages:**

Python, SQL, PySpark, Snowpark.

#### **CI/CD tools:**

Gitlab, AWS Code commit, AWS CI/CD Pipelines

### **Machine Learning:**

Linear and Logistic Regression, Decision Trees, Random Forest, K-Means Clustering, Affinity propagation Clustering, Time Series Forecasting.

### **Deep Learning:**

CNN, RNN, LSTM, Semantic Segmentation, OpenCV, Fasttext, GANs, Reinforcement Learning (DDPG).

### **Business Domains:**

Life Science & Healthcare, Retail and Manufacturing (Open to exploring different business domains)

### **Statistical Methods:**

Hypothesis Testing, Data Correlation, Statistical Analysis, Survival Analysis.



## EDUCATION

Masters of Computer Applications, Vellore Institute of Technology, Vellore  
Specialization: Computer Applications CGPA: 8.7/10

Bachelors of Computer Applications, UOR, Jaipur, Specialization: Computer Applications  
Percentage: 70%

Intermediate - SJS Public School, Raebareli; Percentage- 60%

## 📁 EMPLOYMENT

2022 -  
Present

### **EY, Noida (September 2022 – Present)**

HVAC Spend Analytics Accelerator – Data Scientist

- Spend Analytics Accelerator addresses the insights on different commodity spend by the suppliers.
- As in Data scientist I am responsible for bringing up data from multiple streams such as SAP, HANA and JDE on snowflake and aggregating those on the streamlit dashboard using snowflake python connector
- Development of UNSPSC multiclass classification model using Snowpark UDF for giving better understanding to the commodity managers on the UNSPSC taxonomy.

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| 2021 - 2022 | <p><b>Dunnhumby,Gurugram(June 2021 to Sept 2022)</b></p> <ul style="list-style-type: none"> <li>➤ Offer Pool Recommendations – Railey’s – Data Scientist</li> <li>• Offer Pool recommendation brings best offer to the table which provide maximum profit to the CPGs/retailer</li> <li>• Carry out Reaserch and developement on building offer pools that are profitable to CPGs.</li> <li>• Developing recommendation engine using PySpark, Fasttext, Affinity Propagation Clustering</li> <li>• Review the offer pools recommended with various Stakeholders for validation.</li> </ul><br><ul style="list-style-type: none"> <li>➤ TESCO CE-POS Media– Data Scientist</li> <li>• Analyze various campaigns that are running on TESCO CE Hypermarkets.</li> <li>• Provide significant insight about the campaigns running across Tesco CE- markets</li> <li>• Enhancement of framework that utilizes Pyspark for fetching data from hdfs and uses Robust ANCOVA for providing the insights about the different KPIs for the campaigns.</li> <li>• Showcase the campaign insights with Stakeholders and provide necessay suggesations.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 2018 - 2021 | <p><b>Virtusa,Chennai (June 2018 – May 2021)</b></p> <ul style="list-style-type: none"> <li>➤ Faster AI–SAIL– Machine learning Engineer</li> <li>• Design and Development of No Code AI platform to build Machine Learning and Deep learning models pipelines with just drag and drop.</li> <li>• Build a AutoML Platform starting with Data Ingestion, EDA, Model Training, and Evaluation to showcase the capabilites of SAIL research on Mulitaccuracy and Shapley</li> <li>• Developing of REST APIs and deployment of these APIs on Elastic beanstalk using AWS CI/CD Pipeline.</li> <li>• Log the Machine learning models experiment on MLflow.</li> </ul><br><ul style="list-style-type: none"> <li>➤ PACMAN Ultrasound Image Segmentation-Avanos– Machine learning Engineer</li> <li>• Developing an image segementation Algorithm that segment four anatomical objects (ASM,SCM,MSM and BP) in given Ultrasound Images.</li> <li>• Deployment of on PyQT5 video player in EC2 Windows machine.Pyqt5 player basically shows segmented video of ultrasound along with original video.</li> <li>• It helps client to integrate image segmentation in PACMAN Project.</li> </ul><br><ul style="list-style-type: none"> <li>➤ Tabular Data Synthesizer using GAN.–Medtronics- Machine learning Engineer</li> <li>• Building Streamlit Application aims to simulate synthetic data by using Conditional Tabular GAN.</li> <li>• Also development of Exploratory data analysis in order to validate the quality of original data and synthetic data. Moreover Machine leaning Efficacy tests can be performed to assess the data generated by CTGAN’s.</li> <li>• Project is deployed on AWS infrential instance using Streamlit,S3 ,Pandas,Plotly and Pycaret.</li> </ul><br><ul style="list-style-type: none"> <li>➤ vLife™ AI Applications Marketplace- Machine learning Engineer</li> <li>• Responsible for development of End to End Machine learning and Deep learning solution using EMR/EHR data sources.</li> <li>• Deployment of these models as sandbox environment using Flask REST APIs and AWS CI/CD Pipelines.</li> <li>• Major tech stack used are AWS, Python, Deep Learning, Machine Learning, OpenCV, Reinforcement Learning.</li> </ul><br><ul style="list-style-type: none"> <li>➤ Hospital Recommendation using Reinforcement Learning. - Machine learning Engineer</li> <li>• Refining the existing Recommendation engine using Advance Deep learning methods.</li> <li>•Responsibilites includes development Hospital Recommendation system using Reinforcement learning (Recnn) for Patients based on Hospital Ratings.Solution uses Deep Deterministic policy gradient and MDP algorithm to recommend hospitals/</li> <li>• Project is deployed on AWS EC2 instance using Streamlit</li> </ul> |



## CERTIFICATIONS

IBM Data Science Certified Professional by Coursera

Applied Machine learning with Python Specialization By Michigan University

Deep learning Specialization by Deeplearning.AI

Machine learning A – Z with Python and R by Udemy

Python for Big Data and Spark by Udemy

Statistics for Business Analytics and Data Science by Udemy

Google Cloud Essentials by Linux Academy.



## LANGUAGES

- English
- German (Elementary Proficiency)